

Phase 3 improvement summary

Phase 3 focused on a real limitation in transformed-space zero masking. The repository now includes a data-aware masking fix called `empirical_background` masking.

The current saved Covertypes end-to-end results remain mixed, which means the masking idea is valid but not yet sufficient for a full pipeline win under the current training setup.

The new dataset diagnostics show that Adult Income is a better showcase dataset for the masking improvement itself. Its hidden categorical validity rises from 0.0000 to 1.0000, and its hidden numeric exact-zero rate falls from 1.0000 to 0.0000.

The best next step is to continue from the new notebook and prompt, use ``adult_income`` as the masking-showcase dataset, and expand the Phase 3 workflow to other mixed tabular datasets without overstating the results.

